BILLING AND COLLECTIONS WEB APPLICATION DEV UPDATE

Trece Martires City Water District

Prepared By: **Wally Villanueva**

Reviewed and Approved By:

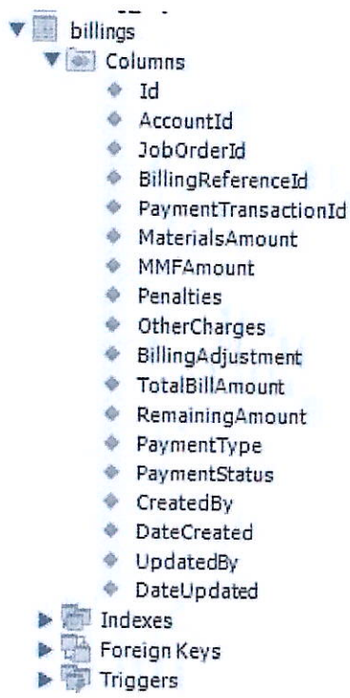
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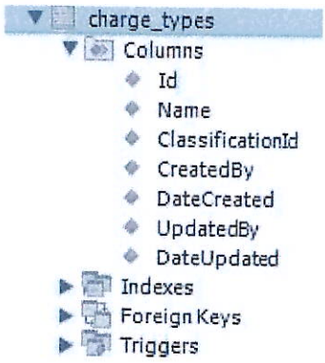
Started with back-end development so that data are readily available for front-end development.

1. **Database:**

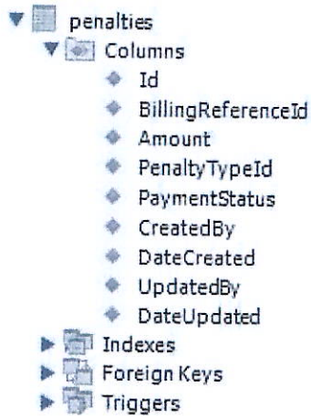
billings table: This table will hold computed billing information like total other charges, penalties and billing adjustments.



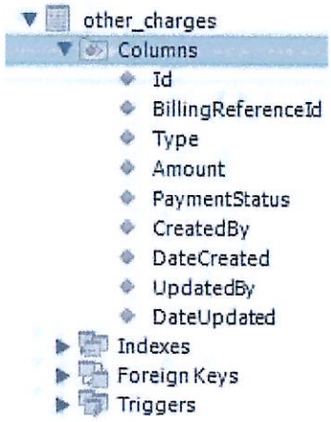
charge_types table: This will serve as a maintenance table for types of charges e.g. Transfer and Widtholding Tax for other charges and unbilled and overdue for penalties.



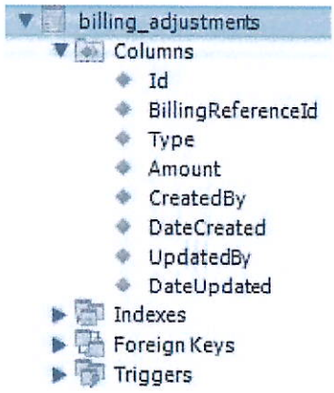
penalties table: Hold account billing charges for penalties e.g. unbilled, and overdue.



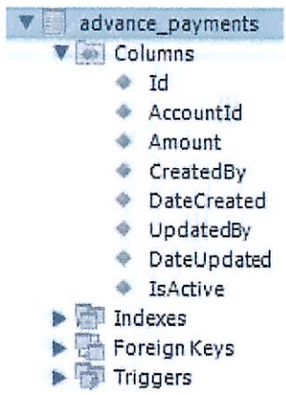
other_charges table: Hold account billing for other charges e.g. transfer and widtholding tax.



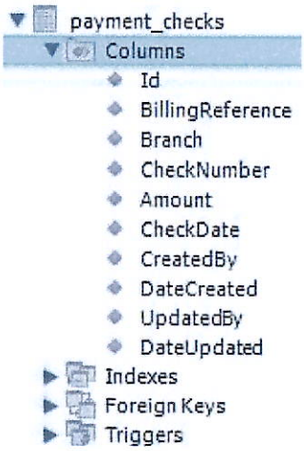
billing_adjustments table: Hold account billing adjustments.



advance_payments table: holds account advance payments.



payment_checks table: Holds information about bank checks used for payment.



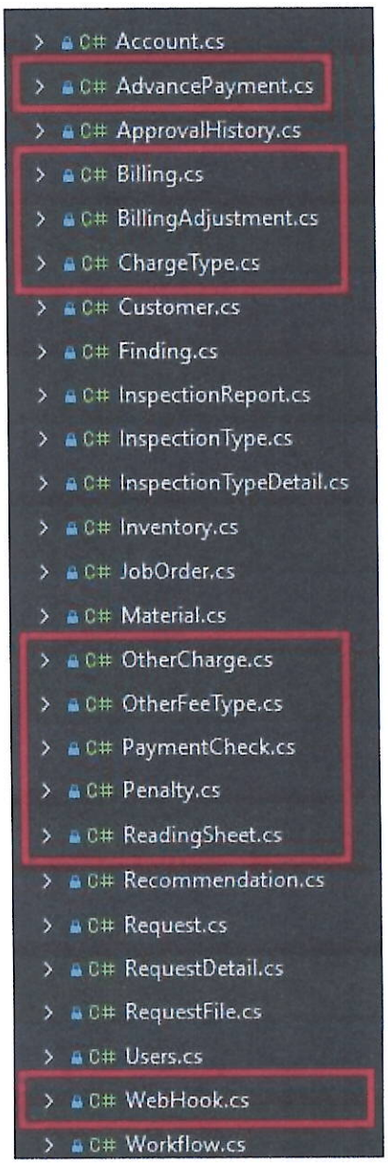
webhooks table: this will hold information of the payment gateway response like gcash, paymaya, and qrph



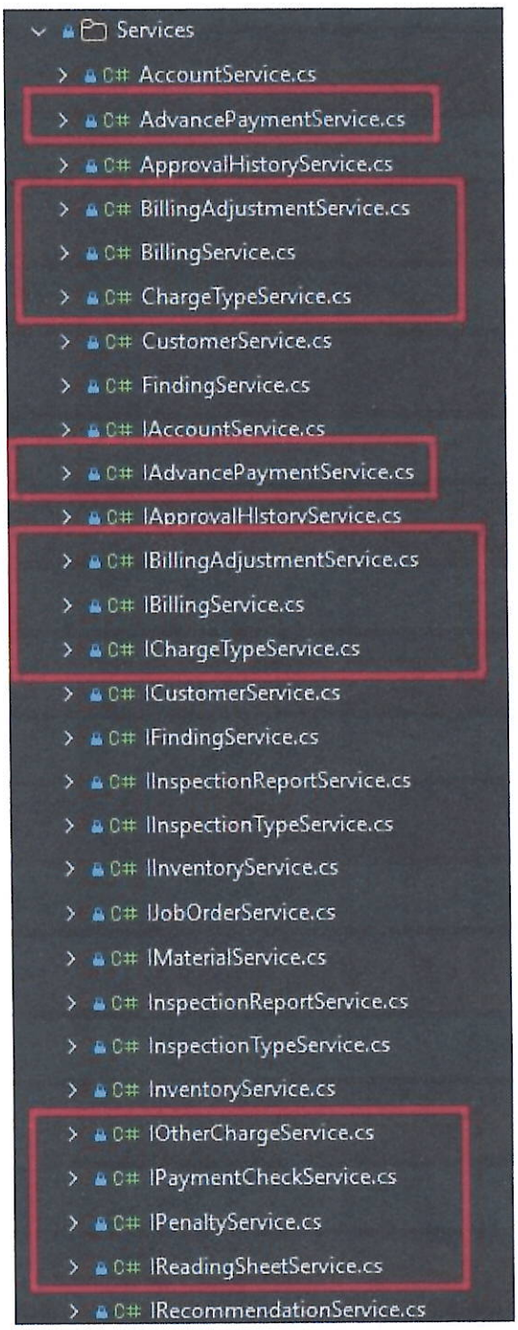
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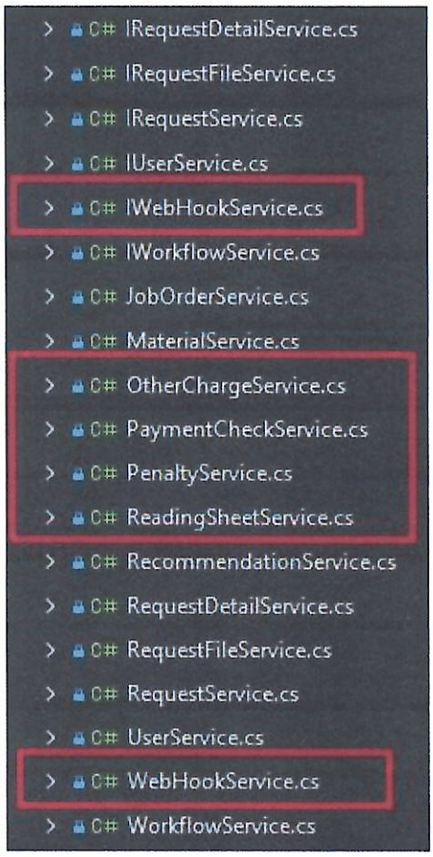
Back-end Code

Data Entities: Data Entities are representation of table and table columns in code for web service.

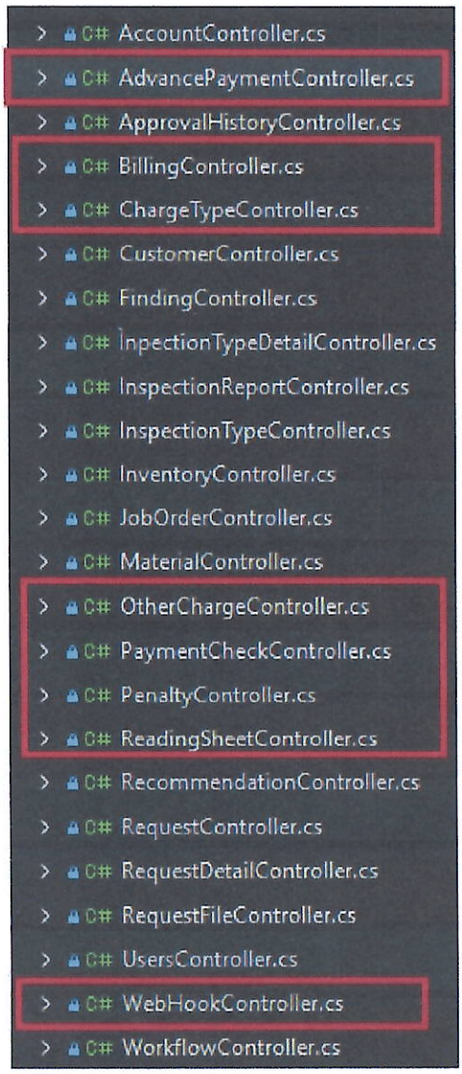


Data Services: This holds the code that will fetch, insert, and update the data from UI to database.

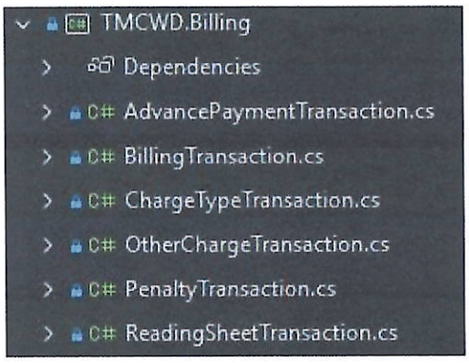




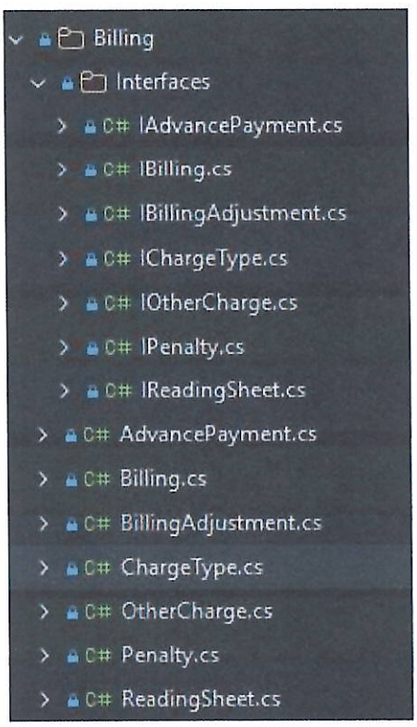
Web Services: this will serve as a data gateway to TMCWD database. We can expose this externally for external application usage like mobile application.



Billing API: this object consumes the web service, contains most of the logic needed by the application, and can also help in creating automated testing.



Models: this is the representation of data table and columns in application level can also help in creating automated testing.



Will continue coding for the list below.

1. **Payments** (Objects within this section will contain the logic for processing payments thru cash, cheque, or e-wallets)
    a. Create database table
    b. Create Data Entity
    c. Add web service objects for data transactions
    d. Add API object for billing and payment processing.
    e. Add codebase that can accept the following payment method (Cash, Cheque, and e-Wallet)
2. **e-Wallet** (Objects within this section will contain logic for processing e-wallet payments using gcash, maya, and other e-wallet platforms)
    a. Create object and codebase for GCash payment to push our payment details to gcash platform or by consuming their SDK (software developer Kit) to directly access gcash payment from our codebase. *(Note: We will be implementing GCash payment only for the meantime. We can add other payment platforms later.)*

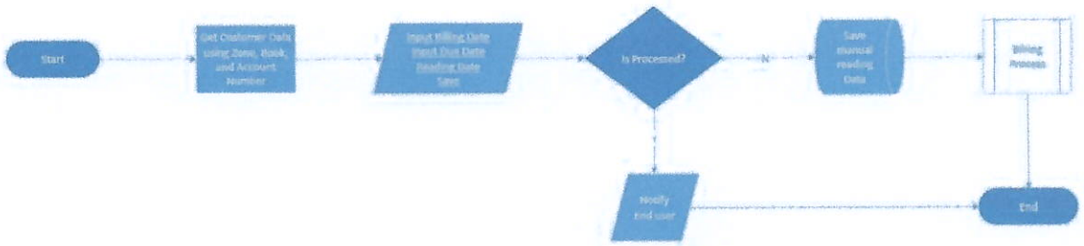
3. **Webhook** (Objects within this section will contain codebase to capture e-wallet response to our system containing details of the payment transaction, we can then update account billing information depending on the e-wallet response to payment transaction we sent and log transactions for future reference.

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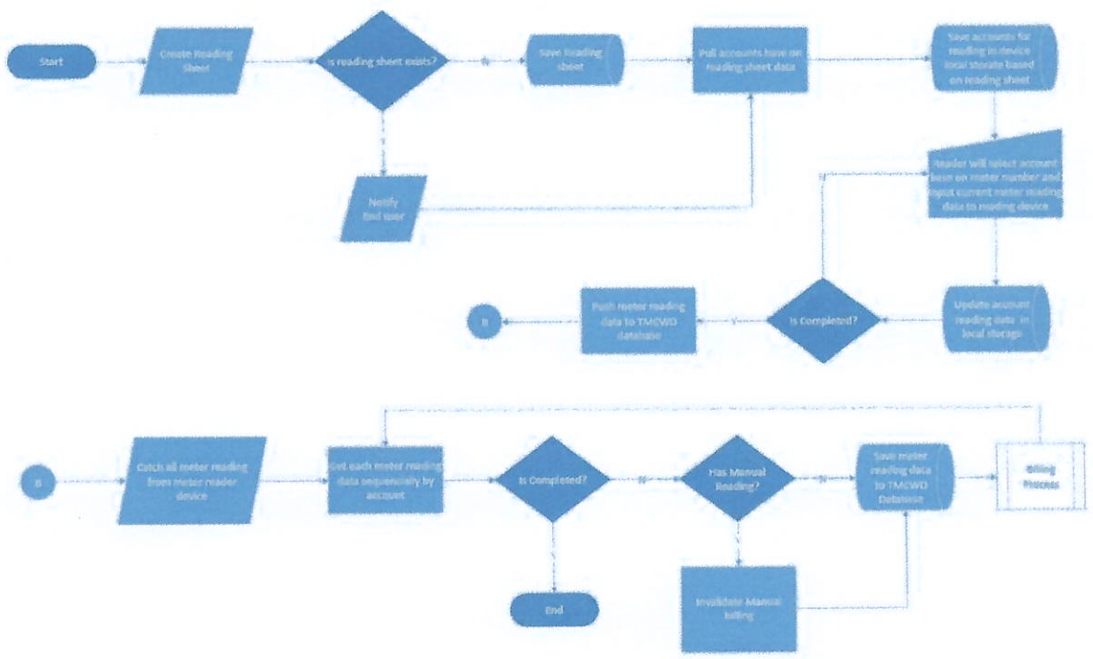
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Read and Bill Flowchart

Manual Reading



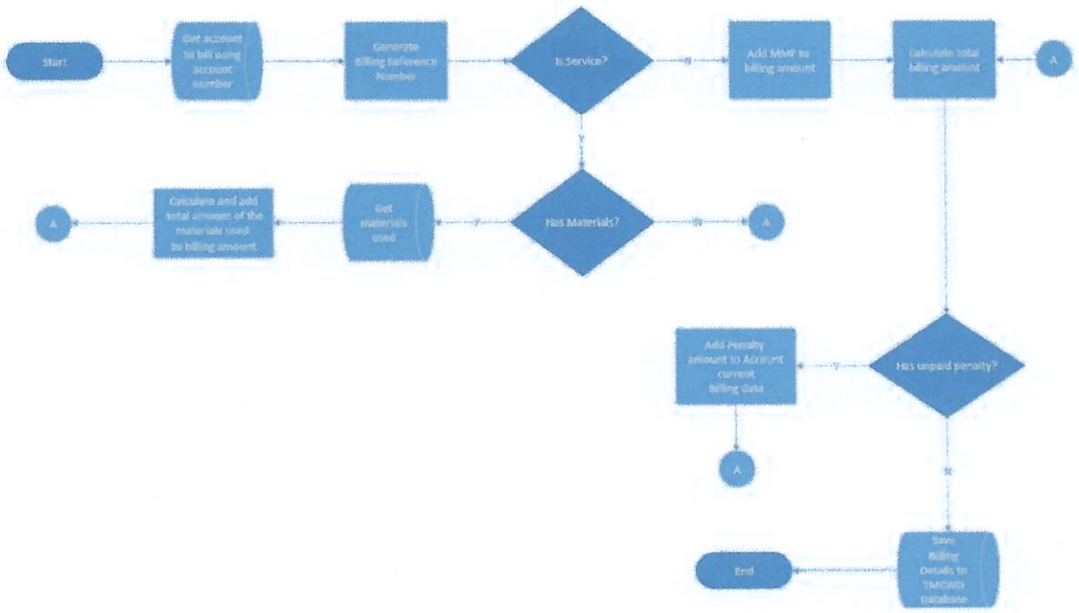
Automated Reading Thru Meter Reader



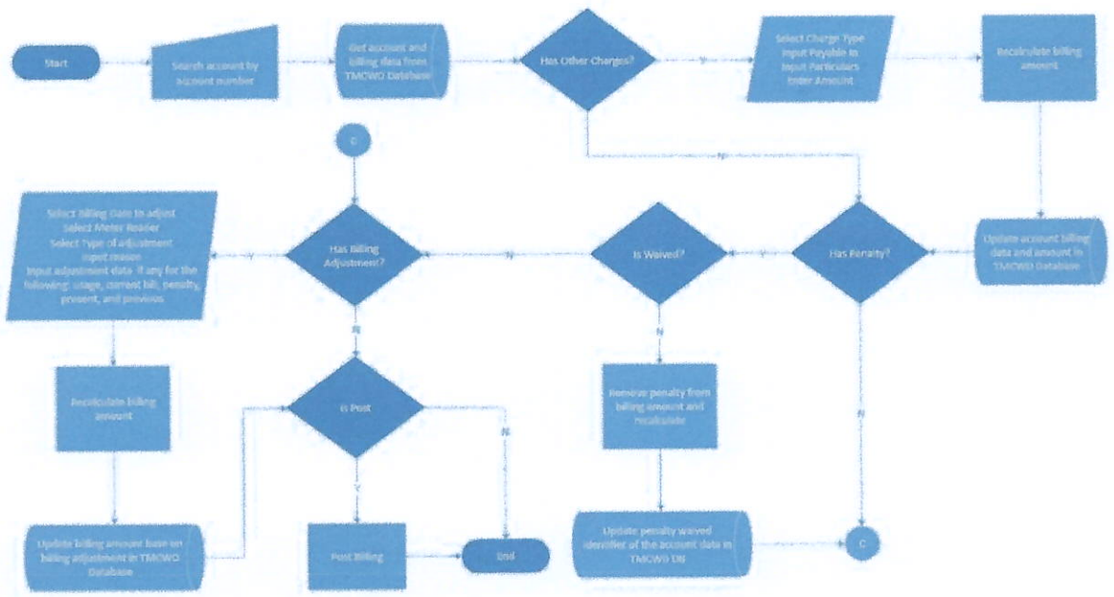
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Billing Process



Additional Billing Transaction



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Payment Application

